
UiPath RPA Internship Training

Email Automation | Integration Service | Exception Handling

Agenda

- ~~Introduction & Hands-on~~

- ~~What Is RPA?~~
- ~~RPA Tools~~
- ~~Installing UiPath~~
- ~~Uipath Automation Methodology~~
- ~~Explore Uipath Studio~~

Day1

- ~~Variables, Data Types ,ControlFlows & Activites~~

- ~~Working with File and Folders, Debugging~~

- ~~Data Manipulation(string,list,Dictionaryes)~~

- ~~Ui Automation~~

- ~~Project 1~~

- ~~Data Manipulation(Data Tables)~~

- ~~Excel & PDF Automation~~

- Email Automation and integration services

- Api Automation + Error handling

Day2

Day3

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Agenda



- Project 2
- UiPath RPA Architecture (Studio, Orchestrator and Robot)
- State Machines
- RE – Framework
- RPA Final Project

Day7

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Day9

01

Email Automation in Studio

Send, retrieve, and filter emails using IMAP, SMTP, Office 365, and Google Workspace activities

02

UiPath Integration Service

Combine API and UI automation with connectors, connections, and event triggers

03

Error & Exception Handling

Build robust workflows with TryCatch, Throw, RetryScope, and ContinueOnError

04

Hands-On & Assignment

Combined practical exercise and take-home assignment for all three modules

01

Email Automation in Studio

Automate emails using UiPath Studio — send, retrieve, and filter messages efficiently



Email Automation Overview & Activity Packages

INPUT

- Names / IDs coming in **subject or body**
- Input files coming as **attachments** (.xls, .pdf)
- Email triggers starting a workflow

OUTPUT

- Progress **reports** to managers
- Exception alerts** and notifications

Key Activity Packages



UiPath.Mail.Activities

SMTP, IMAP, POP3 protocols



Microsoft Office 365

Outlook 365 integration with OAuth



Google Workspace

Gmail integration with OAuth2

Email Protocols at a Glance



SMTP

Sending emails



IMAP

Receiving & managing



POP3

Receiving only



Exchange

Enterprise email

IMAP & SMTP Activities — Properties & Configuration

Get IMAP Mail Messages

Host: imap.gmail.com
Port: 993 (SSL enabled)
Email / Password: Logon credentials
MailFolder: "Inbox"
Top: Max messages to retrieve
OnlyUnreadMessages: True/False
TimeoutMS: Default 30000ms
Output: List of MailMessage

Send SMTP Mail Message

Host: smtp.gmail.com
Port: 465 or 587
From / To / CC / BCC: Recipients
Subject: Email subject line
Body: Plain text or HTML
IsBodyHtml: True for HTML
Attachments: File paths list
SecurePassword: SecureString



Key Tips

Use **SecurePassword** (SecureString) instead of plain Password for security. Enable **SSL/TLS** for encrypted connections. Use **Integration Service** option for cloud-based authentication without storing credentials.

Typical Email Flow



Office 365 & Google Workspace Email Activities



Microsoft Office 365

Authentication:

App ID, Tenant ID, Client Secret

Key Activities:

- Get Mail – retrieve from folder
- Send Mail – with attachments
- Reply to Mail – thread replies
- Move / Delete Mail – manage

Google Workspace (Gmail)

Authentication:

OAuth2 – Client ID, Client Secret

Key Activities:

- Use Gmail – scope activity
- Get IMAP Mail (Gmail)
- Send SMTP Mail (Gmail)
- Supports Integration Service

Feature	Office 365	Google Workspace
Protocol	Microsoft Graph API	IMAP + SMTP
Auth Method	OAuth2 (App Registration)	OAuth2 (Client ID/Secret)
Best For	Enterprise Microsoft shops	Google-centric organizations

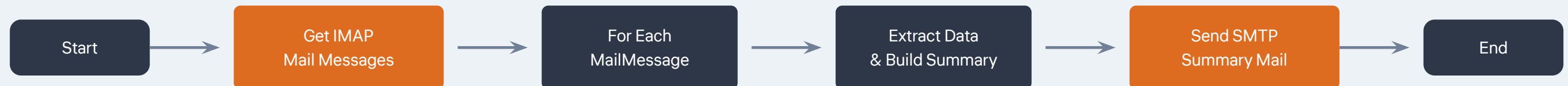
Hands-On — Email Automation Exercise

PRACTICAL

Objective: Build a workflow that retrieves unread emails and sends a summary report

- 1 **Get IMAP Mail Messages**
Server: imap.gmail.com, Port: 993
OnlyUnreadMessages = True, Top = 10
- 2 **For Each Email (TypeArgument: MailMessage)**
Extract: item.Subject, item.Sender
Check: item.Attachments.Any
- 3 **Build Summary String**
Concatenate subject + sender + attachment count for each email into a summary variable
- 4 **Send SMTP Mail Message**
To: manager@company.com
Subject: "Daily Email Summary", Body: summary

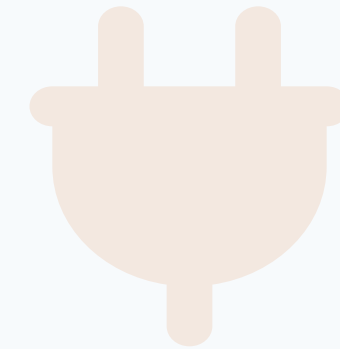
Workflow Diagram



02

UiPath Integration Service

Combine API and UI automation for streamlined development with third-party apps



Integration Service Overview & Three Core Components

Integration Service provides standardized authentication and API access to third-party applications



Connectors

Pre-built API integrations for enterprise apps

- Outlook • Salesforce
- Slack • ServiceNow
- OneDrive • SharePoint



Connections

Authenticated links between UiPath and third-party apps

- User credentials
 - OAuth tokens
- Status: Connected
- Folder-aware



Triggers

Event-based automation starters

- Record Created
- Record Updated
- Record Deleted
- Polling: 5 min default

How It Works

1. Select Connector from the catalog for your app (e.g., Salesforce)

2. Create Connection by authenticating with app credentials

3. Configure Trigger to auto-start workflows on events with data filters

Setting Up Connectors, Connections & Triggers

Creating a Connection

Step 1: Go to **Automation Cloud** > Integration Service > Connectors tab

Step 2: Search and **select your app** (e.g., Microsoft Outlook 365)

Step 3: Click **Add Connection** and authenticate with app credentials

Step 4: Verify status shows **"Connected"** — ready for Studio use

Configuring a Trigger

Step 1: Go to **Triggers** tab > Click **Add Trigger**

Step 2: Select **Connector** and **Connection** from dropdowns

Step 3: Choose **Event** (e.g., "New Email Received") and **Record Type**

Step 4: Select **Process** to execute, **Runtime license**, set polling interval

Setting	Connectors	Connections	Triggers
Location	Integration Service > Catalog	Per-connector page	Integration Service > Triggers
What You Do	Browse and select app	Authenticate with credentials	Select event + process
Result	Pre-built API available	Status: Connected	Auto-starts on event
Key Property	Activities included	Folder-aware sharing	Polling interval (5 min)

Using Integration Service in UiPath Studio

Studio Integration Activities

Package: `UiPath.IntegrationService.Activities`

Use Connection — Scope activity for authenticated operations

Get Records — Retrieve data from connected app

Create Record — Insert new data into connected app

Update Record — Modify existing data

Delete Record — Remove data from connected app

Upload File — Send files to connected storage

Connector Builder



Build **custom connectors** for apps not in the catalog:

1. Import OpenAPI/Swagger definition
2. Configure authentication (OAuth, API Key)
3. Define resources and operations
4. Add triggers with polling URLs
5. Publish to your tenant

Typical Studio Workflow with Integration Service

Use Connection



Get Records



Process Data



Create Record

Close Connection

Activities auto-populate with available connections — no manual API configuration needed

Hands-On — Integration Service Exercise



PRACTICAL

Objective: Create a Gmail connection and Send Email in studio

1

Create Connection in Integration Service

Select Gmail connector, authenticate with you mail, verify Connected status

2

Build Studio Workflow

Use GSuite activities to Send an email

03

Error & Exception Handling

Detect, anticipate, and handle exceptions to build robust and reliable automation workflows



System Exceptions vs Business Exceptions

System Exceptions (Technical)

NullReferenceException

Using a variable with no set value

IndexOutOfRangeException

Index exceeds collection limits

SelectorNotFoundException

UI element not found within timeout

ImageOperationException

Image not found within timeout

ApplicationException

Target application not responding

Business Exceptions (Logic/Data)

Incomplete Data

Required fields missing in input

Invalid Data Format

Data doesn't match expected format

Out of Bounds

Value exceeds allowed limits

Validation Failure

Data fails business rule checks

Human Intervention Required

Robot stops, needs manual review

Key Differences

- **System exceptions** are automatically generated by the runtime — all derive from **System.Exception**
- **Business exceptions** must be explicitly defined by the developer using the **Throw** activity and handled in **TryCatch**

Try Catch, Throw & Rethrow Activities

TryCatch Structure

Try Catch Activity

TRY

Activities to monitor for exceptions. If error occurs, execution jumps to Catches.

CATCHES

Multiple exception-type handlers. Most specific match executes. Can include Log Message, alternative logic, or Rethrow.

FINALLY

Always executes — cleanup, close resources

Throw vs Rethrow



Throw Activity

Creates a **new exception** with a custom message. Used to explicitly raise business rule errors.

Example: `new BusinessException("Invalid invoice amount")`



Rethrow Activity

Propagates the **caught exception** up the call stack, preserving original context. Used after logging or cleanup in Catch.

Use case: Log error, then Rethrow for outer handler



Shortcut: Press **Ctrl+T** to surround any activity with TryCatch

Retry Scope & ContinueOnError Property

Retry Scope Structure

Retry Scope

ACTIONS

Activities to retry on failure

CONDITION

Zero or one activity. False = retry again

NumberOfRetries: Max attempts (default 3)

RetryInterval: Seconds between retries (default 5)

ContinueOnError Property



Type: Boolean (True / False)

Default: False — stops on error

When True: Execution continues despite errors

Warning: If True inside TryCatch, no error is caught

When to Use What — Quick Reference

Scenario	Use This	Why
Intermittent failures	Retry Scope	Auto-retries with delay
Need error details & fallback	TryCatch	Full control over handling
Optional UI element	ContinueOnError=True	Skip if not found

Hands-On — Bank Transaction Validation

PRACTICAL

Objective: Validate bank transactions while handling business and system exceptions

- 1 **Read Excel in TryCatch**
Read BankTransactions.xlsx inside a TryCatch. Catch FileNotFoundException and log an error message
- 2 **Loop & Validate Transactions**
For Each transaction, validate Account Number, Transaction Type, and Amount before processing
- 3 **Throw & Handle Exceptions**
Throw BusinessException for invalid data, simulate a system exception, and handle each type separately
- 4 **Track & Summarize Results**
Keep Success, Skipped, and Failed counters, then display a processing summary at the end

Workflow Diagram



Catch: FileNotFoundException → Log error message

Handle BusinessException & System.Exception separately

Assignment — Employee Registration Email Processor

Task 1: Email Automation

Build a workflow that:

1. Retrieves unread Gmail emails with "**Employee Registration**" in subject
2. Saves the attachment to the Input folder
3. If no attachment, log a warning and skip

Hint: Filter unread + Save Attachment

Task 2: Integration Service

Build a workflow that:

1. Creates a **Gmail connection** in Integration Service
2. Verifies the status is **Connected**
3. Sends the final summary email here

Hint: Summary includes Total, Successful, Failed

Task 3: Exception Handling

Build a workflow that:

1. Reads **employee Excel/CSV data**
2. Validates Name, Dept, Salary; Throw if invalid
3. **Catches** FileNotFoundException, BusinessException, System.Exception

Hint: Track Processed, Successful & Failed counts

INTEGRATION TASK : COMBINE ALL THREE

Build one end-to-end automation that processes registration emails:

- ① Use an **Gmail connection (Integration Service)** to retrieve unread employee registration emails
- ② Save the attachment, then **read the employee file** using TryCatch for error handling
- ③ **Validate each record** and Throw BusinessException for invalid data
- ④ Use **TryCatch** to catch each exception type and keep record counters
- ⑤ Send a **summary email via Gmail** with Total, Successful and Failed records

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Thank You

Continue Your RPA Learning Journey

- Explore more courses on **UiPath Academy**
 - Practice with **real-world scenarios**
 - Build **production-ready** automations
 - Prepare for **UiPath Certification**